

Question ID: 96c3e32d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

One side of a flat board has an area of **874** square inches. If a pressure of **19** pounds per square inch of area is exerted on this side of the board, what is the total force, in pounds, exerted on this side of the board?

Correct Answer: 16606

Rationale

The correct answer is **16,606**. It's given that one side of a flat board has an area of **874** square inches. If a pressure of **19** pounds per square inch of area is exerted on this side of the board, the total force exerted on this side of the board is **(874 square inches)($\frac{19 \text{ pounds}}{1 \text{ square inch}}$)**, or **16,606** pounds.